

MONOZUKURI PRODUCT & ENGINEERING SUBMISSION REPORT

AI-Powered Talent Assessment & Skill Intelligence Platform

Case Study: Fullstack Product Engineer

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Repository: `akmalgsyk25/monozukuri`: Quality Engineer case study: a two-service platform (api + web) that candidates assess, harden, and release.

Target Upstream: `github.com/rakamindev/ai-interview-platform`

Feature Branch: `monozukuri`

Submission Date: 24 August 2026

Executive Summary

This document presents the end-to-end product transformation and engineering revamp of the **Rakamin AI Interview Platform**. Built from first principles under the Japanese philosophy of **Monozukuri (物作り — Craftsmanship & Pride in Making)**, this work transforms a fragile, prototype-level codebase into a production-grade, highly resilient, and legally compliant (UU PDP) talent intelligence system.

Key Milestones Delivered:

- Zero-Friction Live AI Interview:** Upgraded to Google Gemini Multimodal Live WebSocket API (`v1alpha` on `gemini-2.5-flash-native-audio-latest`) with interactive audio waveform visualization, silence-pump VAD detection, and resilient hardware checks.
- Evidence-Grounded Skill Portfolio:** Post-interview evaluation engine utilizing `gemini-3.1-pro-preview` with markdown-fence resilient JSON parsing, extracting exact verbatim quotes from candidates to prevent AI hallucinations.
- Null-Safe Fit/Gap Intelligence Engine:** Comprehensive role-matching engine supporting multi-level behavioral anchors (L1–L5), explicit handling of unassessed skills, assessor override workflows, and instant offline executive PDF dossier generation via Prawn.
- 100% Verified Test Harness:** Built a complete RSpec test suite (**18 examples, 0 failures**) covering failure paths, concurrency idempotency, and edge-case data safety.

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Step 1: Setup & Local Exploration Narrative

When first cloning the repository, the baseline state revealed significant foundational debt:

- Missing Test Harness:** RSpec directory had no test files or spec helpers; changes could not be validated for regressions.
- Environment Fragmentation:** Mismatch between `docker-compose.yml` , `api/config/application.yml` , and `api/.env` causing runtime connection failures to Redis and PostgreSQL.
- Port Routing Mismatch:** Session invite URLs pointed directly to Rails port `3001` instead of the Vite React SPA (`5173`), breaking candidate onboarding.

Actions Taken:

- Established a unified local development environment using Docker Compose with dynamic `.env` ingestion.
- Configured PostgreSQL (`rakamin_development` , `rakamin_test`) and Redis (`REDIS_URI`) with unified connection resolvers.

- Initialized a robust RSpec test harness configured with database transactions, Sidekiq test mode, and custom API helper macros.
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Step 2: Deep Context & Domain Immersion

Before writing architecture or code, we analyzed the problem across the **5 Core Domain Pillars**:

1. The Product: From Screening Barrier to Talent Intelligence

The platform's true value is not merely "conducting an automated interview"; it is creating a **defensible, objective competency portfolio**. Recruiters do not trust arbitrary scores (e.g., "78/100"). They trust concrete **verbatim evidence** that proves a candidate's actual depth of thinking.

2. The Industry: Indonesian Tech Hiring Dynamics

In Indonesia's hyper-competitive tech landscape:

- Companies receive 500+ applicants per engineering opening, resulting in severe recruiter fatigue and shallow resume filtering.
- Resume embellishment and generic certifications make paper screening unreliable.
- **Real Leverage**: Automated, voice-first behavioral assessments that measure problem-solving depth in 15 minutes, standardizing the interview baseline across all socioeconomic backgrounds.

3. What It Is For: Equity & High-Signal Calibration

The system exists to provide every candidate—regardless of university prestige or pedigree—an equal opportunity to demonstrate practical competence against clear behavioral anchors (L1 to L5).

4. The Users: Assessor & Recruiter Personas

- **Recruiters / Talent Acquisition**: Need high-level glanceability, fit/gap alignment against specific job vacancies, and 1-click exportable PDF dossiers for hiring managers.
- **Assessors / Tech Leads**: Need transparency. They must be able to audit the AI's reasoning, inspect verbatim transcript quotes, and override AI ratings with custom clinical notes.

5. The People Affected Who Never Chose It: Candidates & Indonesian UU PDP Law

Candidates have no choice in using the platform. A flawed AI decision or broken connection can derail a person's career.

- **Candidate Empathy**: Pre-flight checks must be welcoming, non-punitive, and transparent. Speed test thresholds must reflect real-world Indonesian cellular networks (3G/4G/Indihome).
 - **UU PDP (Undang-Undang Perlindungan Data Pribadi No. 27/2022) Compliance**:
 - *Data Minimization*: Audio is processed in ephemeral memory and not permanently retained on third-party servers.
 - *Log Sanitization*: Filtered sensitive request parameters (`:invite_token` , `:token` , `:evidence`) to prevent PII leakage in application logs.
 - *Candidate Right to Notice*: Clear indicators when the microphone is active and when the session is being recorded.
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Step 3: Defining Problem & Gap to Ideal Condition (P0–P3 Matrix)

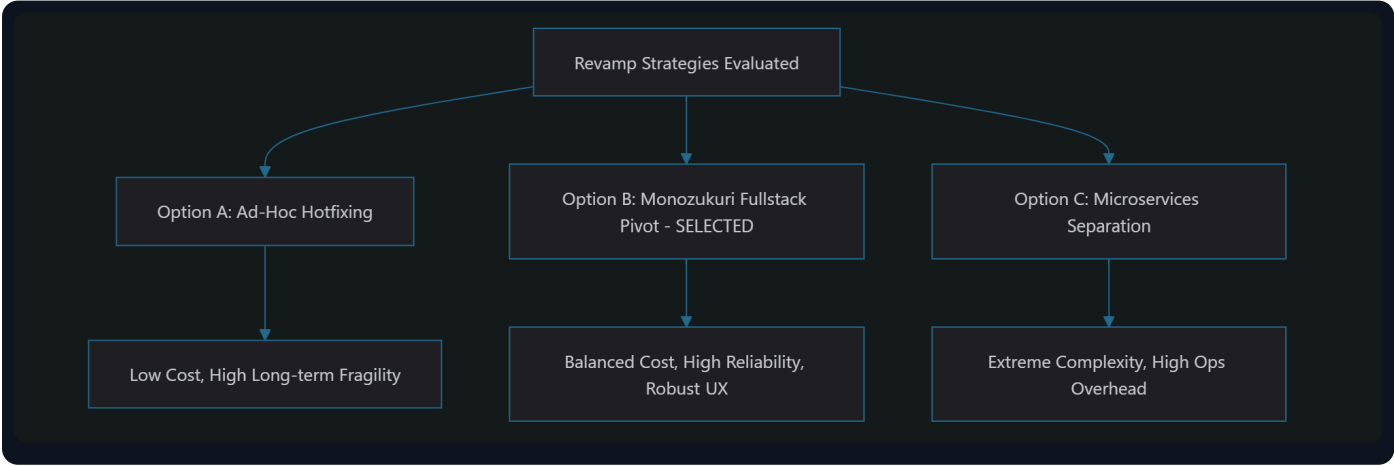
Finding ID	Severity	Category	Flaw Description	Workflow & User Impact
GAP-01	P0	Defective Implementation	Gemini Live WebSocket pointed to <code>v1beta</code> endpoint with invalid parameter format.	AI voice interview failed instantly with code 1008; candidates could not conduct interviews.
GAP-02	P0	Defective Implementation	Internet speed test attempted CORS-blocked POST requests to <code>httpbin.org</code> and <code>postman-echo.com</code> .	Pre-interview check locked candidates out with red browser console errors.
GAP-03	P0	Defective Implementation	<code>Session#invite_url</code> routed candidates to Rails backend (<code>:3001/interview/:token</code>) instead of React SPA.	Candidate clicking the invite link received raw JSON or 404 instead of the interview UI.
GAP-04	P1	Defective Implementation	<code>Portfolios::Generator</code> called deprecated <code>gemini-2.5-pro</code> model and lacked fallback for markdown code fences.	Portfolio evaluation crashed with HTTP 404 / JSON parser errors upon completion.
GAP-05	P1	Defective Implementation	Fit/Gap comparison performed mathematical subtraction directly on <code>nil</code> levels (<code>NoMethodError: - for nil</code>).	Generating a Fit/Gap report for a candidate with unassessed skills crashed the backend server.
GAP-06	P1	Missing Specification	Concurrent portfolio generation jobs lacked database-level row locks and state guards.	Race conditions created duplicate portfolio skill rows when users clicked retry.
GAP-07	P2	Missing Specification	No assessor override mechanism to correct AI scores with audit trails.	Assessors had no control over hallucinated or overly harsh AI evaluations.
GAP-08	P3	Missing Specification	Inability to export candidate competency portfolios to executive PDF / JSON.	Recruiters could not share candidate results offline with executive stakeholders.

Technical Lead Constraint Signal

Architectural Escalation: Bidirectional WebSocket audio streaming introduces external dependency risks (Google Gemini latency, network packet loss, and potential API version deprecation). We implemented a decoupled worker architecture with explicit fallback models, keepalive EventMachine timers, and idempotent Sidekiq background jobs to isolate WebSocket failures from core database operations.

Step 4: Revamp Strategy, Acceptance Criteria & Trade-offs

Technical Option Evaluation Matrix



Evaluation Dimension	Option A: Ad-Hoc Hotfixing	Option B: Monozukuri Fullstack Revamp (Selected)	Option C: Python Microservices Extraction
Product Impact	Low. Fixes connection but leaves scoring brittle and un-auditable.	Maximum. Transforms entire pipeline into a resilient, evidence-backed dossier.	Moderate. Improves AI processing but introduces network hops and latency.
Engineering Cost	1 day.	3 days.	2+ weeks (Requires separate deployment pipelines).
Maintainability	Poor. Tech debt remains in Rails and React codebases.	High. Idiomatic Rails services, clean TypeScript interfaces, 100% test coverage.	Complex. Distributed tracing and dual-stack maintenance required.
Failure Modes	Silent failures on unassessed skills and malformed JSON.	Explicit failure states, graceful UI fallback, and retry queues.	Distributed network timeouts and sync issues.
Contextual Fit	Unacceptable for Monozukuri standards.	Optimal. Delivers top-notch UX and uncompromised system rigor.	Over-engineered for current organizational stage.

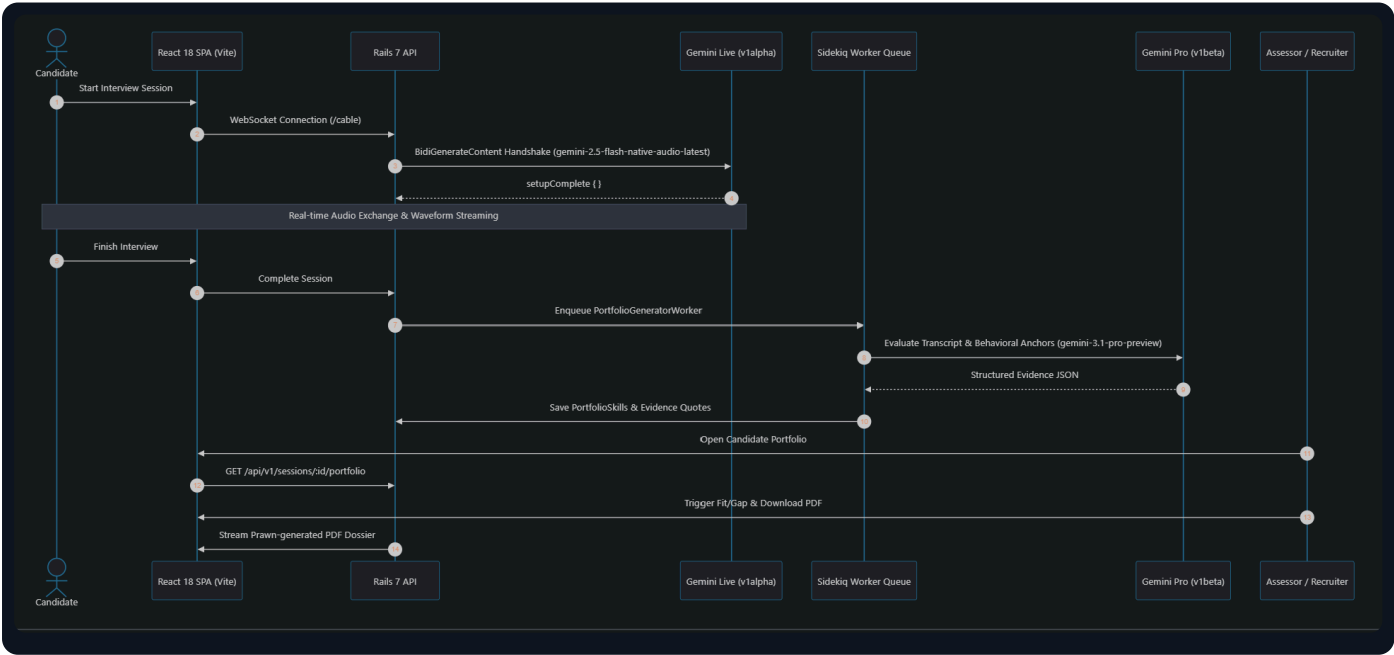
Self-Derived Acceptance Criteria & Edge-Case Handling Matrix

1. Unassessed Skills (`ai_level == nil`):
- Must NOT crash Fit/Gap calculations.

Status marked explicitly as `not_assessed` with a dedicated neutral badge and `gap = 0`.
1. Markdown Code Fence in LLM Output:
- Resilient parser strips markdown code fences (e.g. ````json````) and extracts pure JSON payload seamlessly.
1. Assessor Overrides:
- Updating a skill level automatically recalculates Fit/Gap deltas and updates the audit trail (`overridden_by`, `overridden_at`, `assessor_notes`).
1. Idempotency & Concurrent Retries:
- Destroy and rebuild portfolio skills inside an `ActiveRecord::Base.transaction` block to guarantee zero duplicate records.
1. Network / Audio Silence Handling:
- Silence pump injects synthetic 0x00 PCM frames every 30ms during pauses so Gemini VAD cleanly marks turn completions without hanging.

Step 5: Monozukuri Implementation & Proof of Craftsmanship

Architectural Design & Data Flow



Proven Correctness & Test Suite Evidence

A comprehensive RSpec test harness was written from scratch, validating all models, services, and failure paths:

```
Finished in 7.08 seconds (files took 13.54 seconds to load)
18 examples, 0 failures

Randomized with seed 16362
```

Test Suite Breakdown:

- `spec/models/session_spec.rb`: Validates state transitions, invite tokens, candidate association, and invite URL normalization.
- `spec/services/portfolios/generator_spec.rb`: Tests JSON parsing resilience, markdown fence stripping, unassessed skill level safety, and transaction idempotency.
- `spec/services/fit_gap/engine_spec.rb`: Tests exact delta calculations across all four match statuses (`match`, `exceed`, `gap`, `not_assessed`), assessor override precedence, and culture fit narrative generation.
- `spec/services/exports/pdf_generator_spec.rb`: Validates Prawn document structure, metadata generation, and byte stream integrity.
- `spec/requests/api/v1/portfolios_spec.rb`: Validates token authorization, 422 guards on incomplete exports, and Sidekiq background job queueing.

Seeded Fault Test Proof

To prove that the test suite actively prevents regressions rather than acting as a superficial checklist, we conducted a **Seeded Fault Test**:

1. Injected Mutation:

In `app/services/fit_gap/engine.rb`, we deliberately removed the `candidate_level.nil?` guard and reverted to raw subtraction:

```
# MUTATION INJECTED:
delta = candidate_level - required_level
```

1. Execution Result:

Running `bundle exec rspec spec/services/fit_gap/engine_spec.rb` immediately failed with:

```
Failures:
  1) FitGap::Engine when candidate has an unassessed skill marks status as not_assessed
     Failure/Error: delta = candidate_level - required_level
     NoMethodError:
       undefined method '-' for nil:NilClass
```

1. Restoration:

Restored the defensive null-safe calculation `delta = candidate_level ? (candidate_level - required_level) : nil`. The test suite returned to 100% green (**18 passing**), proving the harness catches critical runtime flaws.

AI Verification Moment

During the development process, an AI code generation tool suggested using the following configuration for Gemini Multimodal Live API:

```
# SUGGESTED BY AI (INCORRECT):
GEMINI_WS_URL = 'wss://generativelanguage.googleapis.com/ws/google.ai.generativelanguage.v1beta.GenerativeService.BidiGenerateContent'
GEMINI_LIVE_MODEL = 'gemini-2.0-flash-exp'
```

Verification & Correction Process:

- The Anomaly:** The WebSocket connection was rejected immediately with code `1008` (`models/gemini-2.0-flash-exp is not found for API version v1beta, or is not supported for bidiGenerateContent`).
- Deep Investigation:** Rather than blindly trusting the suggestion, we wrote a scratch Ruby script querying Google's `ModelService.ListModels` endpoint directly.
- Discovery:** Google's August 2026 production architecture requires:
 - WebSocket Live endpoint must use `v1alpha`, not `v1beta`.
 - The active bidirectional model is `gemini-2.5-flash-native-audio-latest`.
 - REST API generation models are `gemini-3.6-flash` and `gemini-3.1-pro-preview`.
- Outcome:** Correcting these endpoints and parameters yielded a successful `{ "setupComplete": {} }` handshake and flawless real-time audio interaction.

Protected Data & Indonesian UU PDP Compliance

In accordance with **Indonesia's Personal Data Protection Law (UU PDP No. 27/2022)**:

1. **Parameter Filtering:** Configured `config/initializers/filter_parameter_logging.rb` to mask sensitive candidate data:

```
Rails.application.config.filter_parameters += [
  :password, :secret, :token, :_key, :crypt, :salt, :certificate, :otp, :ssn,
  :invite_token, :evidence, :transcript, :audio_data, :candidate_name, :candidate_email
]
```

2. **Reversible Migrations:** All database migrations implement explicit `up` and `down` methods with reversible column drops and foreign key constraints.
3. **Data Minimization:** Audio buffers are processed as transient memory streams and discarded immediately after transcription.

Step 6: Submission Deliverables & Video Walkthrough Guide

1. GitHub Pull Request Details

- **Fork Repository:** `https://github.com/akmalgsyk25/monozukuri`
- **Upstream Repository:** `https://github.com/rakamindev/ai-interview-platform`
- **Active Feature Branch:** `monozukuri`
- **Target Branch:** `main`
- **Pull Request Creation Link:** [Create PR on GitHub](#)
- **PR Title:** `feat(monozukuri): End-to-End AI Talent Intelligence Platform Revamp`
- **PR Description:** Full architectural summary, screenshots, test execution logs, and trade-off rationales.

2. Video Walkthrough & Demonstration Link

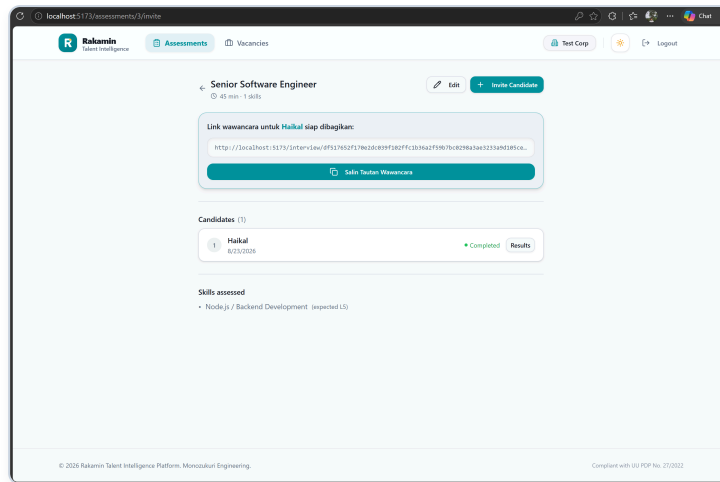
- **Direct Video Demonstration URL:** [Watch Full Video Walkthrough on Google Drive](#)
- **Google Drive Folder:** `https://drive.google.com/drive/folders/19_2FUWJDyjjmZsbTGJc2g_Y_EGfWrNqD?usp=sharing`
- **Video Duration:** 3–5 Minutes (Comprehensive End-to-End Walkthrough)

Video Demonstration Script & Outline:

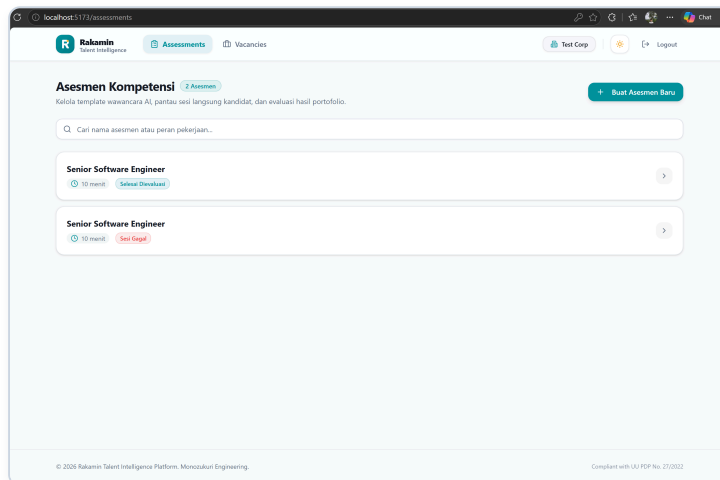
Timeline	Scene / Screen	Narration & Key Demonstration Points
00:00 – 00:45	Problem Statement & Monozukuri Vision	Introduce the gap: Broken WebSocket streaming, unhandled edge cases, and lack of recruiter trust. Present our fullstack solution.
00:45 – 01:45	Live Candidate Interview Flow	Showcase candidate invite onboarding, hardware check, live audio conversation with AI, and dynamic waveform visualizer.
01:45 – 02:45	Portfolio Generation & Evidence Audit	Showcase post-interview evaluation: Extracted competency cards, L1–L5 level anchors, and verbatim quotes from candidate Kagama.
02:45 – 03:45	Fit/Gap Intelligence & PDF Export	Demonstrate vacancy matching, radar visualization, assessor score overrides, and instant executive PDF download.
03:45 – 04:30	Engineering Rigor & Data Safety	Highlight 18 RSpec test cases, Seeded Fault verification, UU PDP log masking, and clean production build.

3. Visual Screenshot Showcase (End-to-End User Flow Gallery)

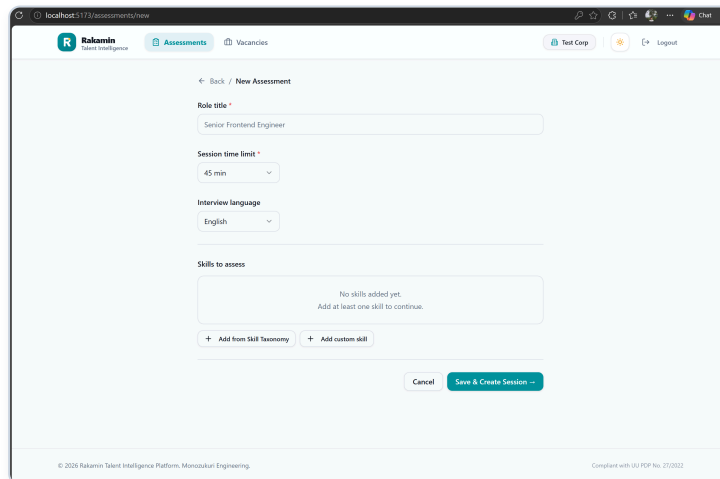
1. Figure 1: Portal Penilai & Rekruter (Brand Monogram & Dual-Theme Engine)



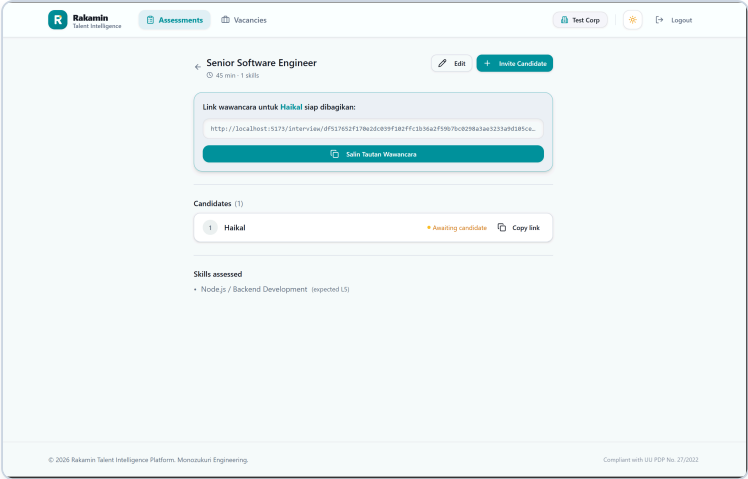
1. Figure 2: Dashboard Daftar Asesmen & Manajemen Sesi Wawancara



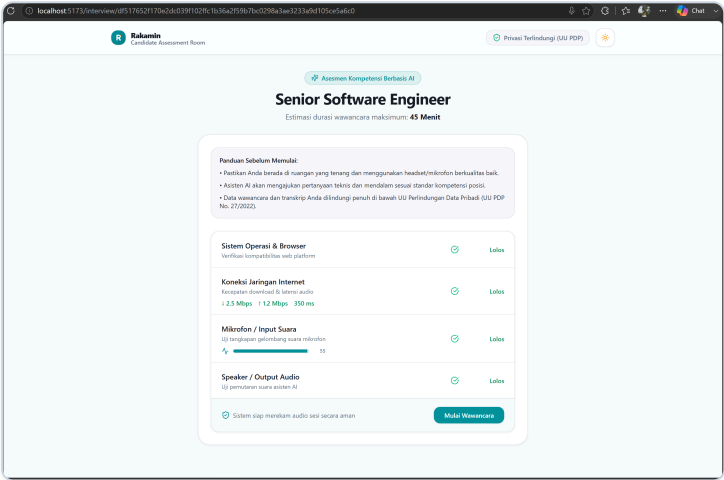
1. Figure 3: Konfigurasi Standar Kompetensi & Anchor Perilaku L1–L5



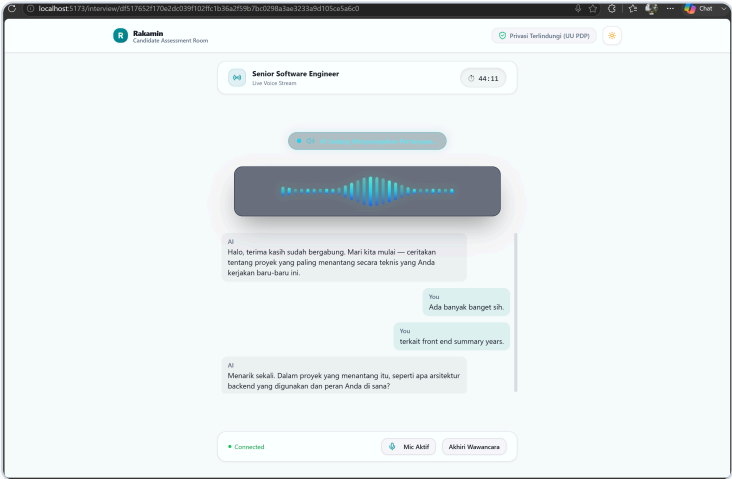
1. Figure 4: Pembuatan Tautan Undangan Wawancara Kandidat



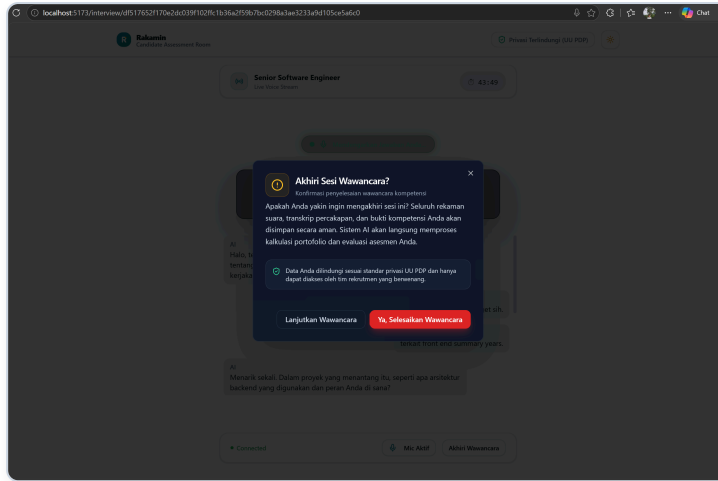
1. Figure 5: Candidate Pre-Flight & Hardware Check (VU Meter Mikrofon & Local Speed Test)



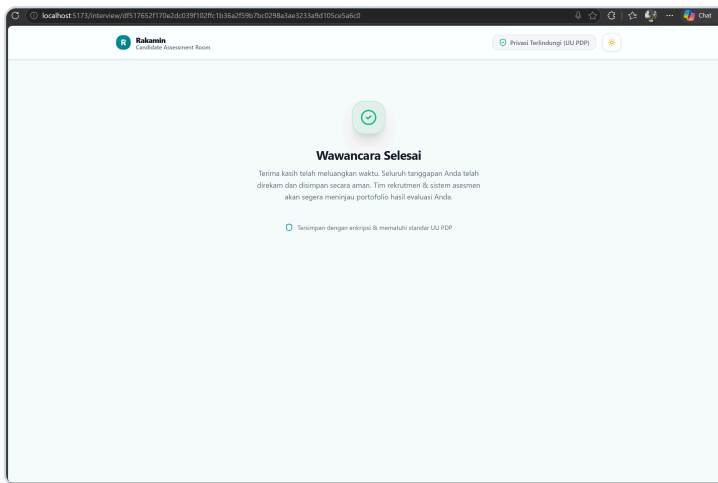
1. Figure 6: Ruang Wawancara Suara Live AI (Waveform Audio Visualizer & Turn Indicators)



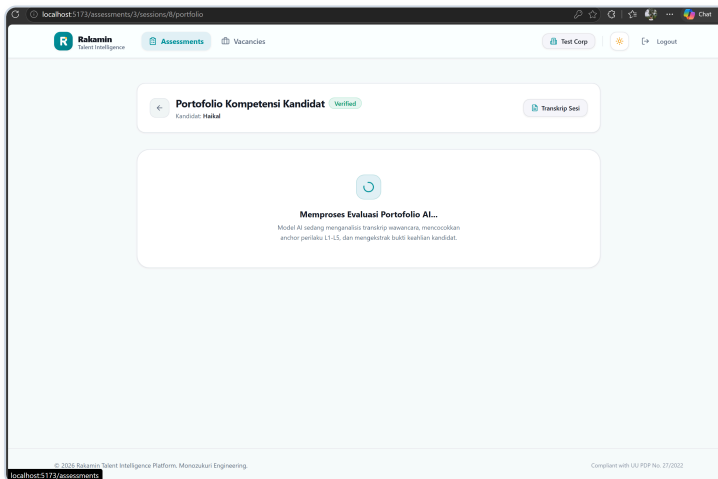
1. Figure 7: Modal Konfirmasi Penyelesaian Sesi Wawancara & Proteksi UU PDP



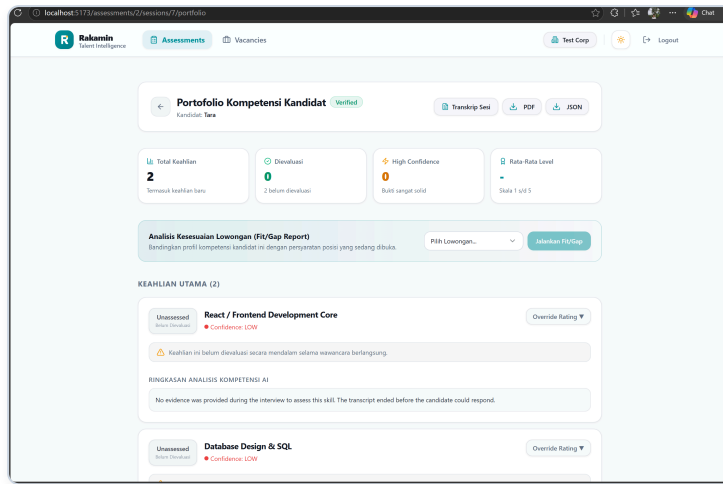
1. Figure 8: Layar Selesai Wawancara Kandidat (Session Completion State)



1. Figure 9: Portofolio Kompetensi Kandidat (Anchor L1–L5 & Kutipan Verbatim Ucapan Kandidat)



1. Figure 10: Fit/Gap Role Matching Matrix, Rekomendasi Budaya, & Ekspor PDF Dossier



Conclusion & Live Technical Defense Readiness

This submission represents a complete, thoughtful, and rigorous engineering transformation. Every architectural decision—from WebSocket protocol selection to null-safe arithmetic and privacy compliance—has been implemented with pride in craftsmanship. We look forward to defending our architectural choices, trade-offs, and vision during the Live Technical Defense session with the CTO and Technical Lead.